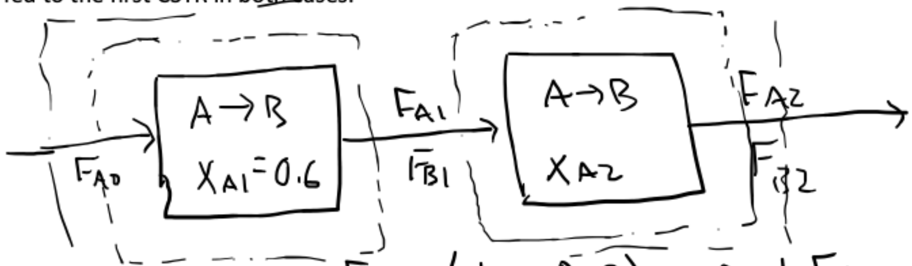
E02 CSTRs in series

The liquid phase reaction A --> B is carried out isothermally and isobarically in a CSTR to 60% conversion. If a second CSTR is added downstream of the first CSTR, what should the volume of that CSTR be to achieve an overall conversion of 90%? Pure A is fed to the first CSTR in both cases.



want $F_{A2} = F_{A0}(1 - 0.9) = 0.1 F_{A0}$

1st CSTR: $V_1 = \dfrac{F_{A0} X_{A1}}{-r_A}$, assume elementary kinetics

$-r_A = k C_{A1} = k \dfrac{F_{A1}}{v}$, $F_{A1} = F_{A0}(1 - X_{A1}) = 0.4 F_{A0}$

$$V_1 = \frac{0.6 F_{A0} v}{k(0.4) F_{A0}} = \frac{2v}{3k}$$

2nd CSTR: $V_2 = \dfrac{F_{A1} X_{A2}}{-r_A}$, $-r_A = k C_{A2} = k \dfrac{F_{A2}}{v}$

$$V_2 = \frac{(0.4) F_{A0} X_{A2} v}{k(0.1) F_{A0}} = \frac{4 X_{A2} v}{k}$$

$$X_{A2} = \frac{F_{A1} - F_{A2}}{F_{A1}} = \frac{0.4 F_{A0} - 0.1 F_{A0}}{0.4 F_{A0}} = 0.75$$

$$V_2 = \frac{3v}{K}$$

$$\frac{V_2}{V_1} = \frac{3v}{K} \cdot \frac{3K}{2v}$$

$$\boxed{V_2 = 1{,}5\,V_1}$$